

```
% loading dataset
T = readtable("airbnbPricing.xlsx", "Sheet", "Sheet1", "ReadVariableNames", true);

namecols = T.Properties.VariableNames; % column names

% few rows of data
head(T);
```

PRICE	DIST	RATE
238	4.06	4.6
138	9.53	4.7
134	7.45	3.6
175	6.19	4
243	1.98	3.7
233	1.98	4.8
251	1.05	4.4
156	8.73	4

Multiple Linear Regression Models

Two Univariate regressions:

```
% first regression model
model1 = fitlm(T, "PRICE ~ DIST");

% second regression model
model2 = fitlm(T, "PRICE ~ RATE");
```

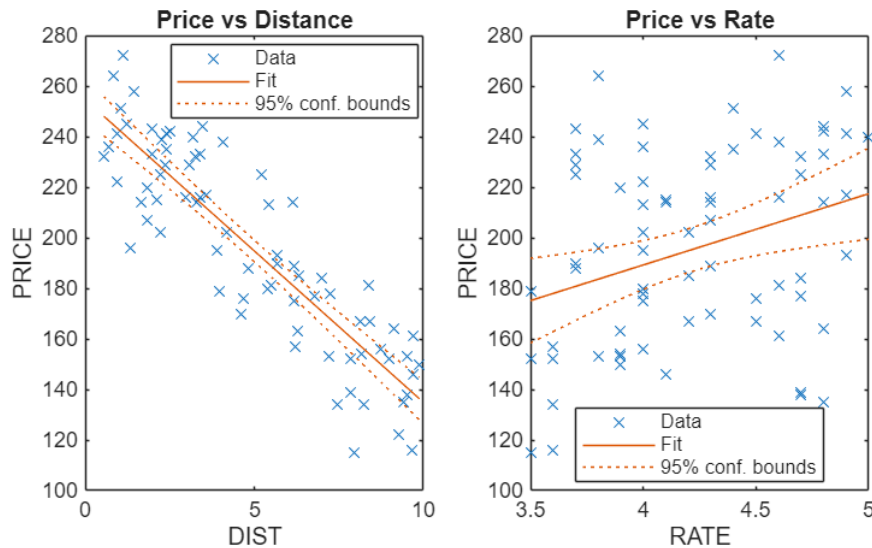
Plotting the two models

```
nr = 1; nc = 2; % no of rows and columns

subplot(nr, nc, 1);
plot(model1);
title('Price vs Distance');
xlabel('DIST');
ylabel('PRICE');

subplot(nr, nc, 2);
plot(model2);
title('Price vs Rate');
xlabel('RATE');
ylabel('PRICE');
sgtitle('Two Univariate Linear Regression Models');
```

Two Univariate Linear Regression Models



Multi-Regression Model

```
modelSpec = 'PRICE ~ DIST + RATE';

mdl = fitlm(T, modelSpec);

% Display the summary of the multiple regression model
disp(mdl);
```

Linear regression model:
PRICE ~ 1 + DIST + RATE

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	177.06	20.041	8.8346	4.2458e-13
DIST	-11.625	0.6808	-17.075	5.0306e-27
RATE	18.008	4.5622	3.9474	0.00018193

Number of observations: 75, Error degrees of freedom: 72
Root Mean Squared Error: 16.9
R-squared: 0.821, Adjusted R-Squared: 0.816
F-statistic vs. constant model: 165, p-value = 1.25e-27

Partial regression coefficient via residual regression (DIST net of RATE)

- Estimate of beta_DIST with univariate regression through the origin

```
inputRes = model2.Residuals.Raw; % input residuals

model3 = fitlm(T, "DIST ~ RATE"); % DIST from RATE

outputRes = model3.Residuals.Raw; % output residuals
```

```

VarNames=["Output residuals" "Input residuals"];

mdlSimpleRegression=fitlm(outputRes,
inputRes,'linear','VarNames',VarNames,'Intercept',false);

% model summary
disp(mdlSimpleRegression);

```

Linear regression model:
Input residuals ~ Output residuals

Estimated Coefficients:

	Estimate	SE	tStat	pValue
Output residuals	-11.625	0.67153	-17.311	9.8291e-28

Number of observations: 75, Error degrees of freedom: 74
Root Mean Squared Error: 16.6

Partial Correlation Matrix

```

TX = table2array(T); % values in a matrix

% partial corr matrix
Rp = partialcorr(TX);

% convert partial corr matrix to a table
RpTable = array2table(Rp, 'VariableNames', namecols, 'RowNames', namecols);
disp(RpTable);

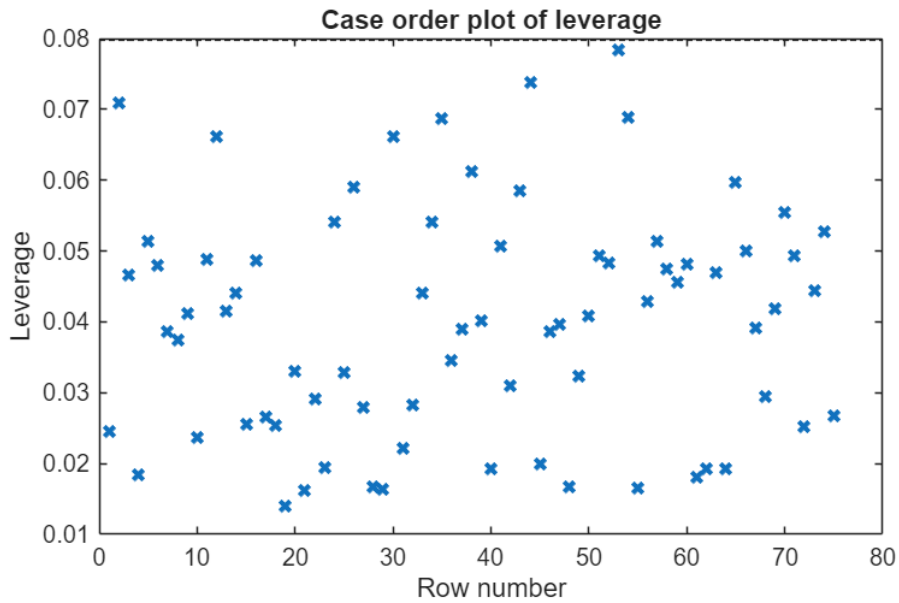
```

	PRICE	DIST	RATE
PRICE	1	-0.89552	0.42179
DIST	-0.89552	1	0.325
RATE	0.42179	0.325	1

```

% Analysis of leverage points
figure
h=plotDiagnostics(mdl,"leverage","LineWidth",2);
h(2).LineWidth=2;

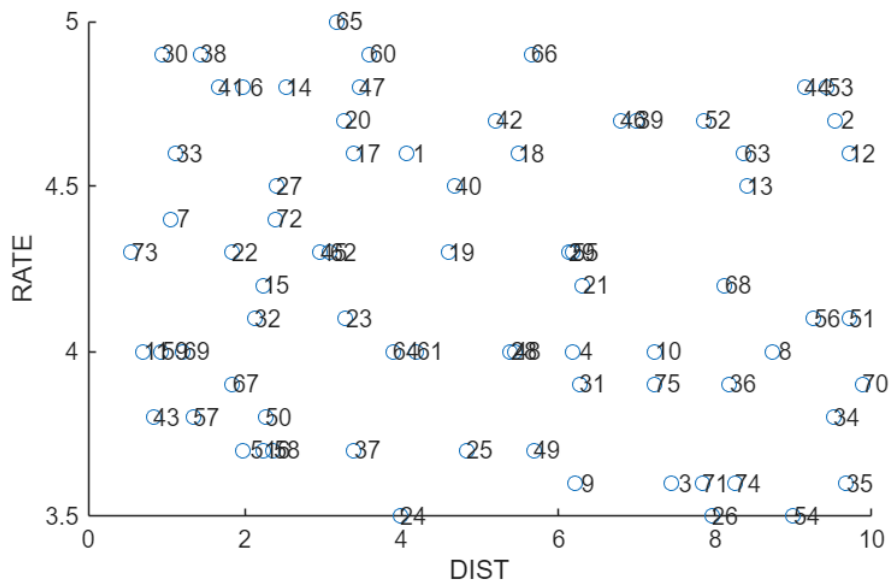
```



Scatter of ASSOR against PROMO

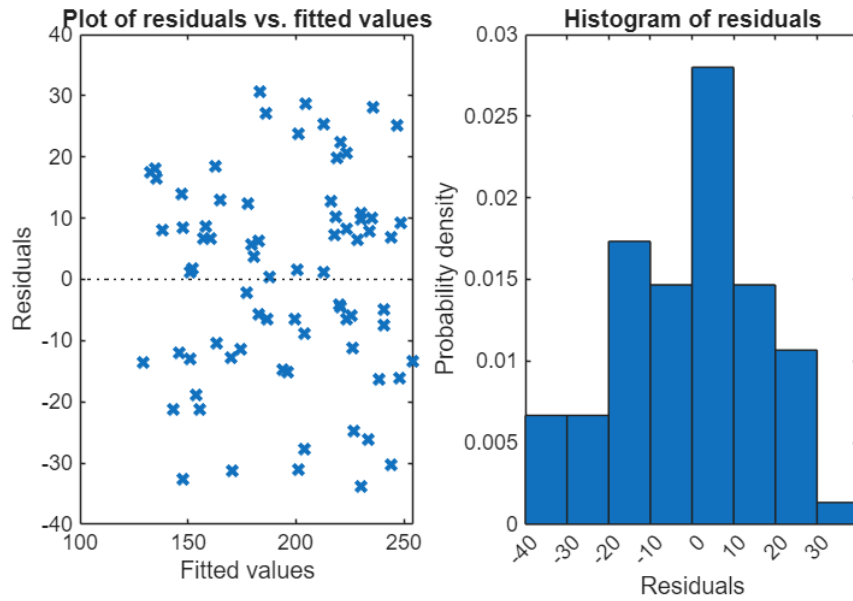
```
figure
n=size(T,1);

scatter(T,"DIST","RATE");
text(T.DIST,T.RATE,num2str((1:n)'));
```



```
subplot(1, 2, 1);
plotResiduals mdl, "fitted", "ResidualType", "raw", "LineWidth", 2);

subplot(1, 2, 2);
plotResiduals(mdl, "histogram");
```



```
x0 = [2.5 4.6];

% prediction
[y_hat0, y_hat_pi] = predict(md1, x0, 'Alpha', 0.01);

disp("Predicted Price for the house, Distance=2.5 KM and Avg Ratings=4.6 is => €" +
num2str(y_hat0));
```

Predicted Price for the house, Distance=2.5 KM and Avg Ratings=4.6 is => €230.8348